



# OpX Accessibility Guideline

*for eLearning Developers*

**2025 Edition**



# Purpose of This Handbook

This handbook is designed to help eLearning developers create inclusive, accessible learning experiences that meet WCAG 2.2 Level AA standards on Articulate Storyline.

## Accessibility Principles

**Perceivable:** Learners must be able to perceive content through at least one sense (sight, hearing, touch).

**Operable:** Learners must be able to interact and navigate using multiple input methods (keyboard, assistive tech).

**Understandable:** Content and navigation must be clear, predictable, and consistent.

**Robust:** The course must work reliably with assistive technologies and across devices.



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# Focus Order

## What It Is

Focus order defines the sequence in which interactive and readable objects are accessed by a keyboard or screen reader. When learners use the Tab key or assistive technology, Storyline moves through objects in this defined order; revealing text, images, buttons, and other elements one by one.

## Why It Matters

A logical focus order ensures that:

Keyboard users can navigate content smoothly and predictably.

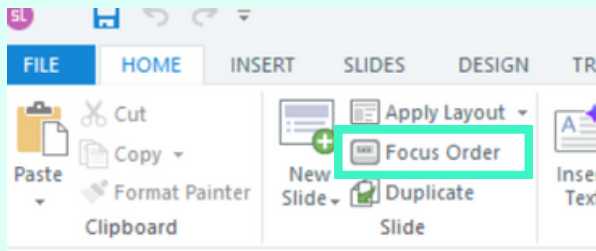
Screen reader users experience content in the same sequence as sighted users.

The course remains usable, understandable, and inclusive for everyone.

Disordered or cluttered focus order can cause confusion, skipped content, or repetition; breaking accessibility compliance and usability.

## Where to Find It in Storyline?

Go to: Home → Focus Order



## Pro Tip

Revisit the focus order after duplicating or copying slides, as Storyline sometimes resets object order or re-adds hidden elements.

## How to Manage Focus Order?

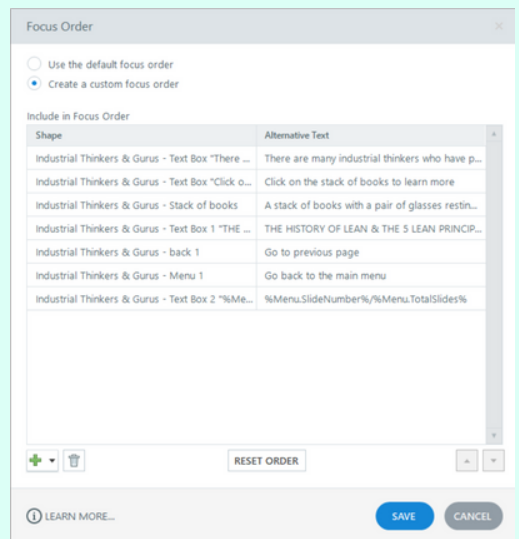
- Set a Logical Sequence
- Arrange items to follow the visual reading order (top-to-bottom, left-to-right).

## Hide Unnecessary Items

- Decorative shapes, background images, or purely visual design elements should be unchecked (hidden from accessibility tools).
- This reduces clutter for screen reader users.

## Check Alternative Text (Alt Text)

- Every visible, meaningful object must have accurate alt text in the Alt Text column.
- Decorative items should have no alt text and be marked as hidden.
- Avoid empty alt text fields for important objects. This results in “blank” reads by screen readers.



Focus Order

☐ Use the default focus order  
☒ Create a custom focus order

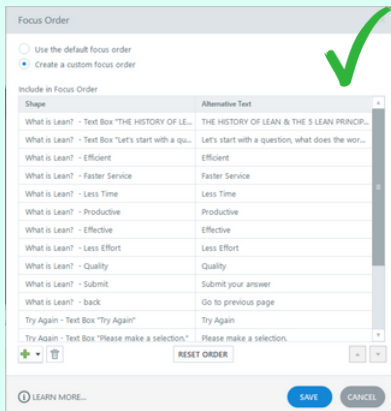
Include in Focus Order

| Shape                                              | Alternative Text                                  |
|----------------------------------------------------|---------------------------------------------------|
| Industrial Thinkers & Gurus - Text Box "There ...  | There are many industrial thinkers who have p...  |
| Industrial Thinkers & Gurus - Text Box "Click o... | Click on the stack of books to learn more         |
| Industrial Thinkers & Gurus - Stack of books       | A stack of books with a pair of glasses restin... |
| Industrial Thinkers & Gurus - Text Box 1 "THE ...  | THE HISTORY OF LEAN & THE 5 LEAN PRINCIP...       |
| Industrial Thinkers & Gurus - back 1               | Go to previous page                               |
| Industrial Thinkers & Gurus - Menu 1               | Go back to the main menu                          |
| Industrial Thinkers & Gurus - Text Box 2 "%Me...   | %MenuSlideNumber%/%Menu.TotalSlides%              |

RESET ORDER

LEARN MORE...

SAVE CANCEL



Focus Order

☐ Use the default focus order  
☒ Create a custom focus order

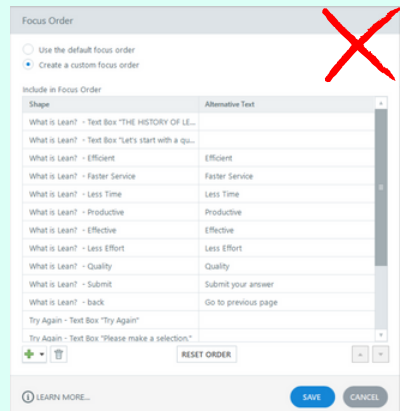
Include in Focus Order

| Shape                                              | Alternative Text                                  |
|----------------------------------------------------|---------------------------------------------------|
| What is Lean? - Text Box "THE HISTORY OF LE...     | THE HISTORY OF LEAN & THE 5 LEAN PRINCIP...       |
| What is Lean? - Text Box "Let's start with a qu... | Let's start with a question, what does the wor... |
| What is Lean? - Efficient                          | Efficient                                         |
| What is Lean? - Faster Service                     | Faster Service                                    |
| What is Lean? - Less Time                          | Less Time                                         |
| What is Lean? - Productive                         | Productive                                        |
| What is Lean? - Effective                          | Effective                                         |
| What is Lean? - Less Effort                        | Less Effort                                       |
| What is Lean? - Quality                            | Quality                                           |
| What is Lean? - Submit                             | Submit your answer                                |
| What is Lean? - back                               | Go to previous page                               |
| Try Again - Text Box "Try Again"                   | Try Again                                         |
| Try Again - Text Box "Please make a selection."    | Please make a selection.                          |

RESET ORDER

LEARN MORE...

SAVE CANCEL



Focus Order

☐ Use the default focus order  
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Include in Focus Order

| Shape                                              | Alternative Text                                  |
|----------------------------------------------------|---------------------------------------------------|
| What is Lean? - Text Box "THE HISTORY OF LE...     | THE HISTORY OF LEAN & THE 5 LEAN PRINCIP...       |
| What is Lean? - Text Box "Let's start with a qu... | Let's start with a question, what does the wor... |
| What is Lean? - Efficient                          | Efficient                                         |
| What is Lean? - Faster Service                     | Faster Service                                    |
| What is Lean? - Less Time                          | Less Time                                         |
| What is Lean? - Productive                         | Productive                                        |
| What is Lean? - Effective                          | Effective                                         |
| What is Lean? - Less Effort                        | Less Effort                                       |
| What is Lean? - Quality                            | Quality                                           |
| What is Lean? - Submit                             | Submit your answer                                |
| What is Lean? - back                               | Go to previous page                               |
| Try Again - Text Box "Try Again"                   | Try Again                                         |
| Try Again - Text Box "Please make a selection."    | Please make a selection.                          |

RESET ORDER

LEARN MORE...

SAVE CANCEL

## Preview and Test

- Use Tab navigation in Preview mode to confirm the flow matches your intent.
- Verify that focus indicators are visible and that all interactive elements are reachable and labelled correctly.



# Alternative Text (Alt-Text)

## What It Is

Alternative text (alt text) is a short written description of an image, shape, or object that explains its meaning or purpose for users who can't see it. Screen readers announce this text aloud, allowing visually impaired learners to understand non-text content.

## Why It Matters

Alt text is critical because it:

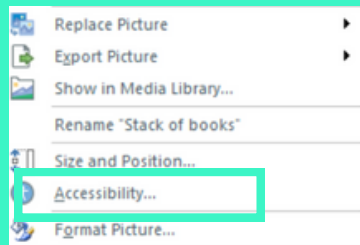
- Ensures equal access to information for screen reader users.
- Helps learners understand visual context that sighted users get instantly.
- Improves overall usability and compliance with WCAG Level A requirements.

Without clear alt text, learners may hear only “graphic” or “button” and lose essential meaning or instructions.

## Where to Find It in Storyline

Select an object → Right-click →  
Accessibility → Alt Text

Alternatively, open the Focus Order panel and type alt text directly in the Alt Text column beside each object.



## How to Write and Apply Alt Text

- Describe Function, Not Appearance
- Explain what the object means or does, not what it looks like.

Example:

✗ “Blue circle with text”

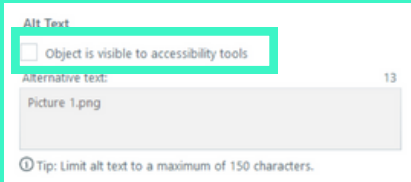
✓ “Next button” or “Continue to next section”

## Keep It Short and Specific

- Aim for one concise sentence (under 125 characters).
- Focus on key information needed to understand the slide.

## Hide Decorative Objects

- Background shapes, design icons, or borders should be unchecked in the Focus Order panel and have no alt text. This prevents unnecessary noise for screen reader users.



## Include Context When Needed

- For infographics or charts, describe the overall meaning, not every visual detail.
- If complex, provide a text version or summary on-screen or in a transcript.

## Buttons and Interactive Elements

- Use the same wording as visible text when possible.
- If a button has an icon only (e.g., a play symbol), write alt text like “Play video.”

## Pro Tip

After writing alt text, test it by turning on NVDA or JAWS and navigating the slide with Tab. If you can understand the purpose of every object without seeing the screen, your alt text is doing its job.



# Providing Alternatives to Non-Accessible Interactivity

Drag-and-drop, sliders, or motion-triggered activities are not accessible.

## Solution Framework

Redesign where possible. Replace dragging with click-to-select, radio buttons or drop-down lists. Feel free to be creative and accessible.

If redesign isn't feasible:

- Add off-screen text conveying the key message or learning point that the activity is meant to communicate.
- Include an off-screen Next button so screen reader users can proceed.
- In the Focus Order, ensure only the descriptive text and Next button are visible to assistive tech.
- Hide non-accessible elements from accessibility tools.
- Test with NVDA or JAWS to ensure the alternative pathway works smoothly.

This activity aims to practice the seven types of waste by matching examples to categories. Here are all the examples with their correct categories:

- Call centre agent without any calls — Waste type: Waiting.
- Asking a customer for the same information twice — Waste type: Over-processing.
- Extra time spent searching for a tool — Waste type: Motion.
- Printing unneeded reports — Waste type: Over-production.
- Sending a customer the wrong appointment date — Waste type: Defects.
- Stocking excessive spare parts — Waste type: Inventory.
- Moving equipment between departments unnecessarily — Waste type: Transportation.

**THE 7 CLASSIC WASTES Activity** The Classic Wastes: TIMWOOD

| Examples                                                      | Categories      |
|---------------------------------------------------------------|-----------------|
| Call Centre agent without any calls                           | Transportation  |
| Asking a customer for the same information more than once     | Inventory       |
| Member of staff walking to a colleague for some information   | Motion          |
| Marketing Materials which are now past their promotional date | Waiting         |
| Sending a customer the wrong appointment date                 | Over-production |
| High levels of customers in a queue                           | Over-processing |
| Moving paperwork between multiple teams                       | Defects         |

Press Space to Access the Keyboard Accessible Version

**Match each of these examples to one of the Seven Wastes**





# Keyboard Navigation and Traps

Ensure all interactions work smoothly with a keyboard and that learners can revisit slides without losing navigation control.

## Checklist

- Every interactive object (buttons, menus, inputs) is reachable via Tab.
- Pressing Shift + Tab moves backward logically.
- No element traps the user, focus can always move forward or back.
- Provide clear visual focus indicators (outline or highlight).
- Avoid using single-key shortcuts that interfere with screen readers.

## Storyline-Specific Navigation Behaviour

- When using feedback layers (e.g., Correct/Incorrect), make sure learners can return to the base layer to use navigation buttons.
- If the feedback layer traps focus or prevents access to main navigation, change the slide property “When revisiting” → “Reset to initial state.”
- This ensures that when the learner revisits the slide, it reloads cleanly and focus resets correctly.
- Use “Resume saved state” only when the learner needs to preserve progress (for example, in multi-step interactions or simulations).

## Pro Tip

Always test revisiting behaviour with Tab navigation to confirm that focus starts at the correct point and navigation remains accessible.

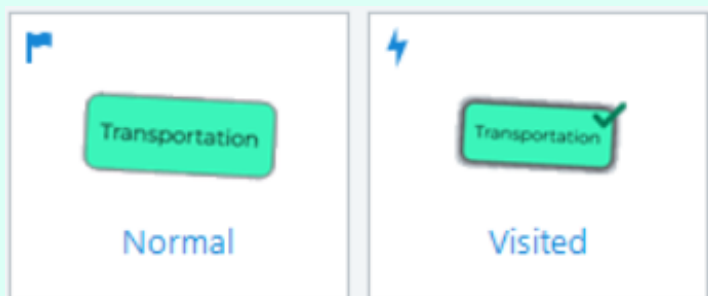
# Colour and Contrast

Provide sufficient visual distinction for low-vision and colour-blind users.

## Guidelines

- Text contrast ratio  $\geq 4.5 : 1$ .
- UI elements (buttons, icons, input fields) contrast  $\geq 3 : 1$ .
- Don't rely on colour alone to convey meaning — combine colour with text, icons, or patterns.
- Use tools like Color Contrast Analyzer, WebAIM Contrast Checker, or Storyline's built-in theme controls.

In the example below, the Visited state of the button is not only shown through outline colour but also reinforced with a check icon, ensuring users who can't perceive colour differences still recognize the change in state.



# Captions, Transcripts, and Audio Descriptions

Make audio-visual content perceivable to everyone.

## Implementation

- Captions for all videos with audio (use Storyline's built-in caption editor).
- Transcripts for audio-only or video-only files.
- Audio descriptions or narrative summaries when key visuals aren't explained in the audio.
- Do not auto-play media, learners should decide when to start playback.
- Always include Play, Pause, Replay, and Volume controls to give full control over media.
- Check timing accuracy and readability of captions.

## Pro Tip

Avoid using looping background videos or sounds that can't be paused. They can overwhelm screen reader output and reduce accessibility.

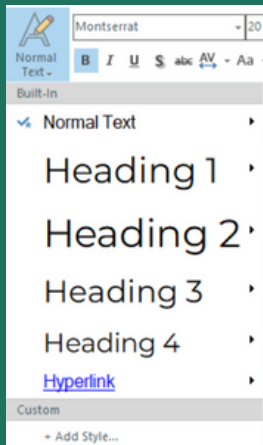


# Text and Layout

## Readable and Flexible Content

- Use true text, avoid embedding text in images.
- Keep text spacing adaptable (avoid fixed-height boxes).
- Don't lock screen orientation unless necessary for activity design.
- Maintain consistent heading styles and logical hierarchy. Use proper heading levels for structure (Heading 1, Heading 2, Heading 3, etc.) and Normal Text for body content. This helps screen readers navigate the content correctly and improves visual consistency.

The image below shows where to find heading and text styles settings. Make sure each paragraph or title uses its appropriate Heading or Normal Text style rather than manual formatting.



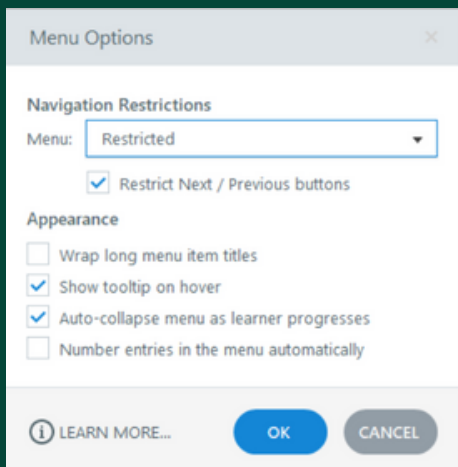
## Pro Tip

Use style labels (Heading 1–4, Normal Text) rather than changing font size or bold manually. This ensures consistent structure for both sighted and screen reader users.

# Navigation and Structure

- Use unique, descriptive slide titles.
- Keep navigation labels consistent throughout (e.g., “Next,” “Back,” “Menu”).
- Ensure menu, support icons or slide number appear in the same location on every slide.
- Offer multiple ways to reach key sections (menu, search, next/previous).

A simple way to do this in Storyline is to use a Restricted Menu setup. This allows learners to move freely between slides they’ve already completed, while still maintaining a guided flow for new content.



Menu Options

Navigation Restrictions

Menu: Restricted

☒ Restrict Next / Previous buttons

Appearance

☐ Wrap long menu item titles

☒ Show tooltip on hover

☒ Auto-collapse menu as learner progresses

☐ Number entries in the menu automatically

[LEARN MORE...](#) **OK** **CANCEL**

Restricting the menu ensures learners can review previous slides without skipping ahead, balancing accessibility and course structure.



# Forms and Input Fields

- Always include visible labels and/or instructions.
- Announce errors clearly — “Please enter your name” — not just “Invalid.”
- Remember previous input values (using Storyline variables) to avoid redundant entry.
- Use descriptive names for form elements in the accessibility settings.

# Motion, Timing, and Animation

- Allow learners to pause, stop, or hide moving or auto-advancing content.
- Avoid flashing or strobing content (> 3 flashes / second).
- If timing is critical (e.g., a timed quiz), provide a way to extend or disable the time limit.
- Never use device motion (tilt, shake) as the only trigger for an action.

# Testing and Validation

| Tool                                     | Purpose                                       |
|------------------------------------------|-----------------------------------------------|
| <b>NVDA / JAWS</b>                       | Test screen-reader experience and focus order |
| <b>Tab Key Navigation</b>                | Validate logical order and traps              |
| <b>Colour Contrast Analyzer / WebAIM</b> | Verify text and UI contrast ratios            |
| <b>Storyline Accessibility Checker</b>   | Run built-in diagnostics                      |
| <b>Zoom Test (200 %)</b>                 | Confirm responsive text and layout behaviour  |

## Recommended Resources

[WCAG 2.2 Guidelines \(W3C\)](#)  
[Articulate Accessibility Guide](#)  
[WebAIM Contrast Checker](#)  
[NVDA Screen Reader](#)